

ADITHYA BHAT

Visa Research, Visa Inc., CA

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SUMMARY

Distributed systems engineer and researcher with 8+ years of experience designing, building, and evaluating fault-tolerant, high-performance distributed systems. Published 13 peer-reviewed papers at top-tier venues (CCS, IEEE S&P, NDSS) and hold 3 US patents. Expertise in building scalable data infrastructure, low-latency protocols, secure distributed systems, and multi-threaded systems in Go, Rust, C/C++, and Python.

TECHNICAL SKILLS

Languages	Go, Rust, C/C++, Python, Java
Distributed Systems	Consensus protocols (BFT/CFT), State Machine Replication, Sharding, Replication, Fault-tolerant systems, Async & Sync protocols
Data Infrastructure	Key-value stores, Distributed caching, Data synchronization, Low-latency protocols, Multi-threading & concurrency, RESTful services
Tools & Frameworks	libp2p, gRPC, Protobuf, Docker, Git, Linux

WORK EXPERIENCE

Visa Research, Foster City (Aug 2023 - Present)
Staff Research Scientist

- Designed and built production-grade distributed systems for payments infrastructure, including an efficient transaction replay system (Ira) and a flexible payment protocol (FPS) for Visa's global network.
- Developed FastSync, a high-throughput data synchronization protocol that reduces sync time from hours to minutes, enabling rapid state recovery across distributed nodes. Filed US patent.
- Architected a partially synchronous sharding protocol and implemented Instachain, a scalable execution system that partitions workload across shards for parallel processing.
- Led research on security and privacy for distributed data systems, including homomorphic encryption for smart contracts and private set intersection protocols.

VMware Research, Remote (May 2021 - Aug 2021)
Research Intern

- Built a prototype token system on VMware's Concord-BFT distributed platform, demonstrating privacy-preserving operations on production-grade distributed infrastructure.
- Designed Quick-Pay, a one-round-trip low-latency protocol optimized for throughput, and extended it into a lock-free sharding solution—eliminating coordination bottlenecks in distributed transaction processing.

Purdue University, West Lafayette (Aug 2018 - Aug 2023)
Graduate Research Assistant – Department of Computer Science

- Designed and implemented multiple fault-tolerant distributed protocols (Apollo, Artemis, Leto, OptRand, RandPiper) handling Byzantine faults in synchronous and asynchronous network settings.
- Built [libnet-rs](#): a high-performance networking library in Rust for distributed system protocols, enabling rapid prototyping and benchmarking at scale.

- Built [libstorage-rs](#) (Rust) and [libe2c-cpp](#) (C++): pluggable storage and consensus frameworks for simulating and evaluating distributed protocols at scale.
- Developed mathematical models for protocol optimization, improving energy efficiency of state machine replication (Middleware 2023). Directly applicable to reducing infrastructure cost at scale.
- Published 10 papers at top-tier venues including CCS (x4), IEEE S\&P, NDSS, and Financial Cryptography. Served on program committees for CCS, FC, and USENIX Security.

EDUCATION

Ph.D., Computer Science – Purdue University, West Lafayette *(2018-2023)*

Advisor: Aniket Kate. Focus: Byzantine fault-tolerant distributed systems and applied cryptography.

B.Tech., Information Technology – NIT Karnataka, Surathkal *(2014-2018)*

SELECTED PUBLICATIONS & PATENTS

13 peer-reviewed publications at CCS (x4), IEEE S&P, DSN, ICDCS, NDSS, Middleware, FC, and Journal of Cryptology. 3 US patents. Selected highlights:

- **EESMR – Energy Efficient State Machine Replication** (Middleware 2023). Optimization models to reduce resource consumption in replicated systems.
- **OptRand – Optimistically Responsive Distributed Randomness** (NDSS 2023). Reconfigurable protocol with optimistic fast-path low-latency performance.
- **Synchronous DKG without Broadcasts** (Journal of Cryptology 2024). Efficient distributed coordination without expensive broadcast primitives.
- **Fast Sync Blockchain System and Method** – US Patent WO2024026321A1. Efficient state synchronization protocol for distributed nodes.

OPEN SOURCE & SOFTWARE ARTIFACTS

- [libnet-rs](#) (Rust) – Core networking library for distributed systems. Powers multiple BFT protocol implementations including Apollo and Sync HotStuff.
- [libstorage-rs](#) (Rust) – Pluggable storage backend library for distributed protocol nodes.
- [libmempool-rs](#) (Rust) – Transaction mempool for buffering, ordering, and managing pending operations in distributed systems.